

Summary

Software Engineering student at 1337 (42 Network) with strong skills in systems programming and backend development. Experienced in C, C++, and TypeScript (NestJS), with projects ranging from Unix tools and real-time servers to containerized infrastructures and RESTful APIs. Passionate about low-level systems, problem-solving, and building reliable software, with a strong desire to learn and contribute in a professional engineering environment.

Projects

- **Transcendence**  **Full-Stack Web Application** *42 Common Core*
TypeScript, NestJS, React, WebSockets, WebRTC, Docker, MySQL
 - Developed a real-time collaborative platform with NestJS, React, TypeScript, and MySQL, integrating WebRTC voice meeting rooms and WebSocket-driven live code editing and chat.
 - Implemented secure JWT & Argon2 authentication, role-based workspace management with transactional ownership integrity, interactive task boards, and document version history.
 - Containerized the entire infrastructure with Docker Compose, including frontend, backend, and database services.
- **Minishell**  **Unix Shell** *42 Common Core*
C, Unix, Processes, Signals, Parsing
 - Developed a fully functional Unix shell from scratch supporting pipes, redirections, environment variables, and built-in commands.
 - Implemented process management with fork/exec, signal handling (Ctrl+C, Ctrl+D, Ctrl+\), and heredoc support.
 - Built a robust tokenizer and parser to handle complex command-line expressions with proper quoting rules.
- **IRC**  **IRC Chat Server** *42 Common Core*
C++, Sockets, TCP/IP, IRC Protocol
 - Built a real-time IRC chat server in C++98 supporting multiple clients over TCP/IP with non-blocking I/O via poll().
 - Implemented operator commands including KICK, INVITE, TOPIC, and MODE with proper permission management.
 - Handled channel management, private messaging, authentication, and full compliance with the IRC protocol.
- **Inception**  **Docker Infrastructure Project** *42 Common Core*
Docker, Docker Compose, NGINX, WordPress, MariaDB, Linux
 - Built a containerized web infrastructure using Docker Compose with separate services for NGINX, WordPress, and MariaDB.
 - Configured custom Dockerfiles, volumes, networks, environment variables, TLS access, and service dependencies on a Linux virtual machine.

Education

- **UM6P/1337 – 42 Network 2024 – 2027**  Ben Guerir, Morocco
Computer Software Engineering; Common Core
 - Built a strong foundation in C/C++, algorithms, data structures, and Unix systems programming through intensive, project-based learning.
 - Developed and debugged low-level and full-stack applications independently in a 100% self-directed, milestone-driven environment.
 - Actively participated in peer-to-peer code reviews and pair programming, continuously refining coding standards, software architecture, and debugging techniques.

Programming Skills

- **Languages:** C, C++, TypeScript, JavaScript, Shell (Bash), SQL
- **Backend & Web:** NestJS, TypeORM, RESTful APIs, WebSockets, WebRTC, JWT Auth & Authorization
- **Databases:** PostgreSQL, MySQL, MariaDB
- **Tools & DevOps:** Docker, Git, GitHub, Linux, Makefile, NGINX
- **Concepts:** Unix processes & signals, Sockets, Memory Management, OOP, Network Programming
- **Spoken Languages:** Arabic (Native), English (Professional Working Proficiency)